

Oleksii Krasnoshtanov

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EDUCATION

Breda University of Applied Sciences

BSc Applied Data Science & Artificial Intelligence; 9.1/10 (4.0 GPA); BUas Scholarship

Breda, Netherlands

Sep 2024 – Present

PROJECTS & TECHNICAL WORK

Measurement Trust Project | *Client project – CEVA Logistics, BUas; team of 4*

Sep 2026 – Jan 2027

- Data engineer on a client-facing project addressing inconsistent item measurement in bol warehouse fulfilment, where size-group misclassification drives fee disputes and repeated seller re-measurement requests

CV Pipeline – Plant Phenotyping Platform | *5-person group project, BUas* | [Code](#) | [Docs](#)

Apr – Jun 2026

- Built an MLOps platform for plant organ segmentation and root-tip detection: U-Net (ResNet-34) behind an authenticated FastAPI backend, with a Next.js frontend, PostgreSQL and Prometheus
- Ran the model lifecycle from Airflow DAGs: data versioning, Azure ML training, drift detection on confidence, latency and error rate, and retraining driven by researcher feedback on flagged predictions
- Owned deployment and CI/CD: one codebase shipping to Docker Compose, on-premise Portainer and Azure Container Apps, with blue-green cutover, health-gated rollback and OIDC cloud auth; 85% coverage gate, 430 tests

Dutch Sign Language Fingerspelling Trainer | *Solo rebuild of a group project* | [Code](#) | [Docs](#)

May – Sep 2026

- Real-time NGT alphabet trainer: MediaPipe landmarks feed a PyTorch residual MLP for the 24 static letters (95% validation) and a bidirectional LSTM over 30-frame sequences for the two needing motion (92%)
- Sole author: FastAPI WebSocket backend streaming to a Next.js 16 / TypeScript frontend, bilingual EN/NL, containerised with CI and published documentation

NPEC Plant Phenotyping System | *Individual Project, BUas* | [Code](#)

Nov 2025 – Jan 2026

- Ranked **#2 of 77** in Kaggle competition for plant root segmentation; selected to present results
- Built the full autonomous loop – U-Net root segmentation, root-tip extraction, pixel-to-robot transform and OT-2 pipette control in PyBullet – benchmarking PID against a PPO reinforcement-learning controller

Emotion Timeline | *Solo rewrite of a group project* | [Code](#)

Sep 2026

- Solo rebuild turning a 51-minute Russian documentary into a per-scene emotion timeline: 316 transcript segments grouped into 47 scenes, each scored by two models trained on different languages
- Built the pipeline behind the classifier: speech-to-text system selection, a training set assembled from datasets with no single language/class overlap, and error analysis on where the two models disagree

HACKATHONS & COMPETITIONS

BrabantHack 2026 | *Winner – Demcon Deep Tech Track* | [Code](#)

Apr 2026

- Won the Demcon track with a model that locates a pedestrian standing entirely outside the camera frame from the shadow they cast into it: 0.626 mean IoU on the organisers' hidden test set; team of 3
- Drove the gain by re-describing the box relative to the frame edge it hides behind – side, distance, width, height, y-centre – on a ResNet-50 trunk fused with 19 shadow descriptors: 0.468 to 0.610 mean IoU, reproducible from the repository

HERE Technologies Hackathon | *RDM Verification Challenge*

May 2026

- Built XGBoost classification pipeline for Restricted Driving Manoeuvre verification combining probe GPS features with Mapillary and OSM geospatial signals; designed synthetic data scrambling for NDA-compliant delivery

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript

ML / Deep Learning: PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers

Computer Vision: OpenCV, U-Net, MediaPipe, PyBullet

NLP & Audio: LangChain, Faster-Whisper, FAISS, Ollama, Gradio

MLOps & Cloud: Azure ML, Azure Container Apps, Airflow, MLflow, Prometheus, Trivy

Infrastructure: Docker, Docker Compose, Portainer, GitHub Actions CI/CD, uv, Ruff, pytest

Backend & Data: FastAPI, Pydantic, PostgreSQL, SQLAlchemy, Alembic, Pandas, NumPy

Frontend: Next.js, React, Tailwind CSS, WebSockets

Spoken Languages: English (fluent), Ukrainian (native), Russian (fluent), Polish (basic)